

Oral Health Status and Oral Healthcare System in Indonesia: A Narrative Review

Fania Chairunisa^{1,2,*}, Aulia Ramadhani^{1,3,*}, Sachiko Takehara¹, Kaung Myat Thwin¹, Tin Zar Tun^{1,4}, Hikaru Okubo¹, Lisdrianto Hanindriyo², Taufan Bramantoro³, Hiroshi Ogawa¹

¹Division of Preventive Dentistry, Faculty of Dentistry & Graduate School of Medical and Dental Sciences, WHO Collaborating Centre for Translation of Oral Health Science, Niigata University, Niigata, Japan, ²Department of Preventive and Community Dentistry, Faculty of Dentistry, Universitas Gadjah Mada, Yogyakarta, ³Department of Dental Public Health, Faculty of Dental Medicine, Universitas Airlangga, Surabaya, Indonesia, ⁴Department of Paediatric Dentistry, University of Dental Medicine, Yangon, Myanmar

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ABSTRACT

Aim: To assess the current situation regarding oral health status and system in Indonesia through the existing literature and government documents, identify challenges, and propose recommendations for the future. **Materials and Methods:** A literature search was conducted from June to October 2023 through the scientific databases of PubMed, Web of Science, and Google Scholar, including published articles from 2013 to 2023. The following main keywords were used: “Indonesia,” “oral health system,” “oral healthcare system,” “oral health policy,” “oral health promotion,” “oral health coverage,” and “oral health status.” Articles without any restrictions on study design, review articles, regional or national oral health reports, and Indonesian health laws or regulations were included. Data extracted from the included studies were synthesized and presented into several subgroups. **Results:** Despite the establishment of a health system in 1982, the oral health status of Indonesian people remains unsatisfying. The prevalence of untreated caries and periodontal disease is still high. Several policies and programs have been implemented to tackle those problems; however, Indonesia still has many challenges to overcome, including geographic constraints, uneven distribution of dental personnel, low public awareness and information on oral care utilization, insufficient resource allocation, and lack of prevention and promotion programs. **Conclusion:** The oral health status of the Indonesian population remains unsatisfactory. Future actions should emphasize on equal access and distribution of dental care, strengthen monitoring and evaluation of ongoing programs, integrate oral health into the national health agenda, and allocate adequate resources to support oral health initiatives.

KEYWORDS: Indonesia, oral health policy, oral health promotion, oral health status, oral health system, universal health coverage

INTRODUCTION

Oral health is as equally important as general health since it reflects an individual's quality of life, mental health, and physical wellbeing.^[1] However, many global populations, particularly in underdeveloped countries, still suffer from oral diseases. Possible reasons for these conditions include limited access to oral health services due to uneven dental workforce distribution, lack

of dental insurance and knowledge, and insufficient

Address for correspondence: Dr. Kaung Myat Thwin,

Division of Preventive Dentistry, Faculty of Dentistry & Graduate School of Medical and Dental Sciences, WHO Collaborating Centre for Translation of Oral Health Science, Niigata University, 2-5274, Gakkocho-dori, Chuo-ku, Niigata 951-8514, Japan.

E-mail: kaung@dent.niigata-u.ac.jp

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*These authors were contributed equally as the first author.

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political attention to integrating oral health into basic health services.^[2,3]

Indonesia initiated achieving universal health coverage (UHC) in 2014 through the National Health Insurance scheme (NHI), which currently covers 95.7% of the total population.^[4] However, the implementation faces challenges, primarily due to disparities in healthcare provision across different areas and inadequate information accessibility, particularly for rural and low-income individuals.^[5]

A comprehensive understanding of Indonesia's oral healthcare system and UHC is essential for delivering individual quality oral care, aligning with the WHO's recommendations for achieving equality and UHC for all.^[6] While some studies have documented Indonesia's oral health status and system, studies addressing its recent situation, particularly after the NHI implementation and its impact on oral health system, are few.^[2,7,8] To bridge this gap, we conducted a review to assess the current situation, identify challenges, and propose future actions for enhancing oral health in the country.

MATERIALS AND METHODS

STUDY DESIGN

This narrative review study focuses on the state of oral health status and system in Indonesia over the past decade.

SEARCH METHODS

The literature review was conducted from June to October 2023 by searching databases such as PubMed, Web of Science, and Google Scholar, including published articles from 2013 to 2023. Due to the limited number of peer-reviewed articles elucidating Indonesia's oral health system, supplementary information was gathered from relevant governmental and non-governmental organizations' websites and documents. The key phrases and Boolean logic were utilized for database searching, including the search terms "Indonesia," "oral health system," "oral healthcare system," "oral health policy," "oral health promotion," "oral health coverage," "oral health," "dental caries," "periodontitis," "periodontal health," "gum disease," "gingivitis," "oral cancer," "oral habits," and "oral condition." These search terms were derived from the PICO criteria: Population (P) constituted the Indonesian population regardless of age; Intervention (I) encompassed changes, programs, or policies related to oral health implemented in Indonesia over the past decade; Control (C) denoted the oral health status and oral healthcare system in 2013; and Outcome (O)

referred to the current situation of oral health status and oral healthcare system in 2023, including changes over the past decade.

INCLUSION CRITERIA

We included articles without any restrictions on study design, review articles, regional or national oral health reports, and Indonesian health laws or regulations. Articles depicting the Indonesian population were selected irrespective of age, gender, or conditions. Additionally, articles featuring oral health systems and/or oral health as a primary focus were considered. The language was limited to English or Indonesian based on resource availability.

EXCLUSION CRITERIA

Commentary articles, letters to the editor, conference presentations, articles that did not comprehensively address oral health status or systems, and those without full texts were excluded.

STUDY SELECTION

AR and FC conducted a literature search for articles in Indonesian language, while KT, TT, HO, and ST were responsible for searching English articles. Initially, all retrieved articles were imported into EndNote, and duplicates were eliminated. Articles underwent screening based on their title and abstracts, followed by a thorough assessment of full-text articles for eligibility. Final studies meeting the criteria were included for further analysis. Two reviewers (AR and FC) independently performed the screening process. Any reviewer discrepancies were resolved through discussion with LH and TB until a consensus was reached.

DATA EXTRACTION AND SYNTHESIS

We utilized a spreadsheet to extract the information from selected articles. Titles, abstracts, and contents were reviewed following the PICO and inclusion criteria. Detailed information regarding the oral health status and systems was collected and extracted as follows: the prevalence of oral diseases (dental caries, gingivitis, periodontitis, tooth loss, oral lesion, and oral cancer), description of oral health programs, oral health system policies and regulations, dental health workforce, NHI and UHC, and dental curricula. All results were reviewed and discussed within the team. We also conducted a narrative synthesis to summarize and describe the current findings. The included articles were categorized to distinguish oral health status and oral health systems before clarifying the present condition and highlighting the challenges. Further analyses such as sensitivity and subgroup analyses

Table 1: List of included studies

Ref. No.	Author	Title	Year	Contents	Type of document
9	Indonesian Health Research and Development Council.	Indonesian Basic Health Survey (RISKESDAS).	2019	Oral health status, dental caries status, and periodontal status.	National report
23	Indonesian Ministry of Health.	Indonesian Health Profile.	2021	Oral health workforce, and dental education system.	National report
24	Indonesian Ministry of Health.	Decree of Indonesian Ministry of Health no HK.01.07/Menkes/671/2020.	2020	Standard for dental therapist professionals.	Regulation
25	Indonesian Ministry of Research, Technology, and Higher Education (RISTEKDIKTI).	Faculty of Dentistry database in Indonesia.	2020	List of Dental Schools in Indonesia.	Online database
26	Indonesian Ministry of Research, Technology, and Higher Education (RISTEKDIKTI).	Guidelines for Medical and Dental Students Profession Competency Examination (Rule No. 18).	2015	Dental student's professional education timeline.	Regulation
27	Indonesian Ministry of Health.	Indonesian Internship Program for Medical Doctors and Dentists (Rule No. 39).	2017	Internship program for fresh-graduate dental students.	Regulation
28	President of the Republic of Indonesia	Presidential Regulation No. 72 of 2012 about the National Health System.	2012	Dental education system.	Regulation
29	World Health Organization	Primary Health Care System (PRIMASYS): case study from Indonesia, abridge version.	2017	Indonesian National Health System.	Report
30	Mahendranta <i>et al.</i>	The Republic of Indonesia Health System Review.	2017	Oral healthcare system.	Electronic books
31	Betri <i>et al.</i>	The Role of Public Health Center (Puskesmas) as The Gatekeeper of National Health Insurance	2016	The role of Public Health Centers in the oral healthcare system.	Journal article
32	Werdhani <i>et al.</i>	Medical problem in Asia pacific and ways to solve it: The roles of primary care/ family physician (Indonesia Xperience).	2019	Oral healthcare system.	Journal article
33	Prabhakaran <i>et al.</i>	Financial Sustainability of Indonesia's Jaminan Kesehatan Nasional: Performance, Prospects, and Policy Options.	2019	Oral healthcare system.	Electronic books
34	The World Bank	GDP (current US\$)-Indonesia	2021	National Health Insurance System.	Online database
35	Indonesian Ministry of Health	Dental and Oral Health Maintenance for Maternal and Child: A Guideline for health personnel at health service facilities	2012	Oral health promotion program.	Government document
36	Indonesian Ministry of Health	A Guideline of Usaha Kesehatan Gigi Sekolah (UKGS)	2012	Oral health promotion program guidelines.	Government document
37	Indonesian Ministry of Health	Regulation of Indonesian Ministry of Health No. 89	2015	Oral health promotion program.	Regulation
38	Indonesian National Social Security Board (BPJS)	A Practical Handbook of Health Service	2015	Oral health promotion program.	Government document
39	Indonesian Ministry of Health	Regulation of Indonesian Ministry of Health No. 67	2015	Oral health promotion program.	Regulation

were not performed due to the heterogeneity of the data collected.

RESULTS

This narrative review, utilizing available literature, government documents, and regulations, examines

Indonesia's oral health status and system from 2013 to 2023 [Table 1]. The analysis reveals that the oral health system and the regulations for oral health promotion programs were integrated and developed in 2015. However, the evaluation of both the system and the programs has not been adequately standardized. As

Table 2: Dental caries status in Indonesia (2013 to 2023)

Study	Study year	Study area	Age group (years)	Sample size	Caries prevalence (%)	DMFT/dmft
Indonesian Basic Health Survey (<i>Riskesdas</i>)	2013	Nationwide	12	23,598	42.6	1.4
			15	19,509	44.3	1.8
			18	16,295	45.2	3.8
			35–44	160,649	61.4	5.4
			45–54	130,997	61.2	7.9
			55–64	81,875	57.0	12.3
			65+	56,783	46.8	18.9
	2018	Nationwide (34 provinces)	3–4 (dmft)	-	81.5	6.2
			5 (dmft)	-	90.2	8.1
			5–9	-	92.6	0.7
			12	-	72.0	1.9
			15	-	68.5	2.4
			35–44	7009	92.2	6.9
			65+	2618	95.0	16.8
	2014	Jakarta	5 (dmft)	390	90	7.5
			12	458	84	3.2
Iswanto <i>et al.</i> , 2016	2016	Tomohon	13–15	30	-	4.3
Maharani <i>et al.</i> , 2019	2016	Jakarta	12	696	-	1.6
Agustanti <i>et al.</i> , 2018	2017–2018	Jakarta	12	487	-	1.7
Khairinisa <i>et al.</i> , 2023	2019	Jakarta	5 (dmft)	266	-	7.7
Andayani <i>et al.</i> , 2021	2019	Lampung	10–12	410	75.6	2.6
Amelinda <i>et al.</i> , 2022	2019–2020	Kaliwates	35–44	97	-	7.8
Marthinu <i>et al.</i> , 2020	2020	Manado	20–50	47	-	6.3
Bahar <i>et al.</i> , 2021	2020–2021	Jakarta	12	139	-	2.3

All DMFT/dmft values are presented as DMFT values unless otherwise specified in parentheses as (dmft). DMFT = decayed, missing, filled permanent teeth, dmft = decayed, missing, filled primary teeth, - = no data available

a result, the current oral health status, which should reflect the outcomes of the oral health system and promotion programs, is not sufficiently representative.

ORAL HEALTH STATUS

According to data from the latest Indonesian Basic Health Research in 2018, dental caries and periodontal diseases are the most prevalent oral diseases in Indonesia.^[9] Dental caries exhibits a high prevalence, affecting 88.8% of the population across all age groups, while periodontal disease affects 74.1% of the total population. Despite 57.6% of individuals experiencing oral health issues, only 10.2% seek treatment for their dental concerns.

Dental caries status

Table 2 shows information on the prevalence of dental caries in Indonesia, drawn from published articles and reports from 2013 to 2023.^[10–18] The overall prevalence of caries is notably high in Indonesia, particularly among older adults aged 65 and above, followed by young children aged 5–9 years. Analysis of *Riskesdas* data from 2013 to 2018 reveals a shifting trend, with a decreasing trend in the MT score from 17.05 to

10.1, while an increasing trend in the DT score from 1.84 to 6.6 was observed in this older age group [Supplementary Table 1]. An increment in the DMFT index was observed across all age groups within 5 years, particularly in DT.

Periodontal status

Table 3 presents the distribution of periodontal status in Indonesia over a 10-year period.^[9,19–22] Although periodontal examination records were not available in the *Riskesdas* 2013 data, periodontal examinations were conducted using the CPI criteria in the latest nationwide survey, *Riskesdas* 2018. The highest prevalence of periodontal disease was observed in the 45-to-54-year age group (77.8%), followed by the 35-to-44-year age group (77.0%). On average, there are more teeth with healthy periodontal tissue than those affected by periodontal disease.

ORAL HEALTH WORKFORCE

In Indonesia, dental care is delivered through a collaborative approach; dentists provide patient treatment, while dental therapists focus on providing maintenance care for the population. As of December

Table 3: Periodontal status in Indonesia (2013 to 2023)

Study	Study year	Study area	Age group (years)	Sample size	CPITN amount (%)				
					Healthy	Bleeding	Calculus	Shallow pocket	Deep pocket
Suling <i>et al.</i> , 2013	2013	Manado	15–16	226	38 (16.8)	6 (2.7)	169 (74.8)	13 (5.7)	0
Susanto <i>et al.</i> , 2020	2016	Bandung	11–14	4	0	1 (25)	3 (75)	0	0
			15–24	115	0	1 (0.9)	81 (70.4)	33 (28.7)	0
			25–34	128	0	0	68 (53.1)	58 (45.3)	2 (1.6)
			35–44	81	0	2 (2.5)	36 (44.4)	42 (51.9)	1 (1.2)
			45–54	47	0	0	23 (48.9)	23 (48.9)	1 (2.1)
			55–64	47	0	0	6 (33.3)	11 (61.1)	1 (5.6)
			65–74	7	0	4 (1)	0	6 (85.7)	1 (14.3)
Susanto <i>et al.</i> , 2018	2018	Bandung	22–34	39	0	0	0	39 (100)	0
			35–44	43	0	0	0	42 (97.6)	1 (2.4)
			45–54	20	0	0	0	20 (100)	0
			>55	22	0	0	0	19 (86.3)	3 (13.7)
Sabrinadevi <i>et al.</i> , 2021	2021	Bandung	19–25	130	0	11 (8.4)	114 (87.6)	5 (3.8)	0
Study	Study year	Study area	Age group (years)	Sample size	Average number of teeth with periodontal disease				
					Healthy	Bleeding	Calculus	Shallow pocket	Deep pocket
Indonesian Basic Health Survey (Riskesmas)	2018	Nationwide (34 provinces)	15–24	-	2.5	0.69	-	0.2	0.01
			25–34	-	3.1	0.74	-	0.4	0.02
			35–44	-	2.7	0.76	-	0.5	0.03
			45–54	-	2.1	0.76	-	0.4	0.04
			55–64	7009	1.4	0.75	-	0.4	0.04
			65+	2618	2.5	0.73	-	0.2	0.02

CPITN = the community periodontal index of treatment needs, - = no data available.

2021, the total number of dentists registered in the Indonesian Medical Council was 33,652 dentists and 4483 specialist dentists [Supplementary Table 2].^[23] Despite the number of dentists increasing significantly each year, there is an issue with the uneven spread of the dentist workforce across the country [Supplementary Figure 1]. Most dentists (22,603) are concentrated in major urban centers, particularly on Java Island, while regions such as Maluku and North Maluku have few (202) dentists.^[24]

DENTAL EDUCATION SYSTEM

In Indonesia, there are 17 public and 15 private dental universities.^[25] These institutions annually produce around 2000–3000 dentists.^[23] The dental program typically takes 5–6 years, culminating in a national dental examination and the acquisition of a dentist license. However, since 2023, newly graduated dentists will be required to participate in an additional 6-month compulsory internship program to deepen their competencies in conducting dental treatment and to help with the shortage of dental personnel in particular areas.^[26,27]

Moreover, there are 18 public and five private dental therapist education programs, each lasting for 3–4 years. These programs could graduate approximately 1000–1500 skilled and proficient therapists annually. Dental therapists must obtain a license from the national government, ensuring adherence to professional standards and regulations.^[28]

ORAL HEALTHCARE SYSTEM

In 1982, Indonesia established its first national health system, which has evolved through subsequent years due to political shifts and regional autonomy. The current system, outlined in the 2012 Presidential Regulation, includes both public and private financing sources and healthcare providers. Private providers manage private healthcare, while public healthcare is overseen by central and regional governments [Table 4].^[29] The Ministry of Health sets national standards and regulations while also supervising some healthcare services. Provincial governments are responsible for the healthcare service and system in their area, whereas district or municipality governments mostly manage primary healthcare services.^[30]

Table 4: Healthcare and delivery system in Indonesia

Public	Authority	Municipality health office	District and/or Provincial government/health office		MoH and Provincial government/health office	
	Responsibilities	Manage public and private healthcare facilities within the area.	Monitoring, supervision, and managing district health services, and coordinate any health issues within or across area.		National standard setting, developing regulations and future direction, ensuring financial and human resources, supervising tertiary and subspecialist hospitals.	
	Healthcare facilities	Community health centers (<i>Puskesmas</i>) and their auxiliary centers.	Hospitals (Class C & D)		Hospitals (Class A & B)	
Referral System		Primary healthcare	↔	Secondary healthcare	↔	Tertiary healthcare
Private	Authority	Networks of hospitals, clinics managed by non-profit organizations, for profit providers, individual doctors/dentists/midwives				
	Healthcare facilities	Primary care clinic, individual doctor/ dentist/midwife, nursing and midwifery clinic	Hospitals (Class C & D) and specialist clinics		Hospitals (Class B) and subspecialist clinics	
	Responsibilities	Provide services for individuals and population, coordinate and report any health issues to <i>Puskesmas</i> or provincial/district health offices.				

Sources: Presidential Regulation No 72 of 2012 about the National Health System (2012); Primary Health Care System (PRIMASYS); case study from Indonesia, abridged version. WHO Geneva; 2017.^[28]

Hospital classifications: Class A: Hospital equipped with broad facilities and medical services. Class B: Hospital equipped with medical facilities and services of at least 11 limited specialists and subspecialists. Class C: Hospital with adequate medical facilities and services of essential specialists. Class D: Hospital with required medical facilities and services. MoH: Ministry of Health

Primary healthcare facilities include community health centers and private providers who play a vital role in maintaining oral health for individuals and communities.^[31,32] Both types of providers may refer patients to secondary and tertiary healthcare facilities for complex dental treatments provided by public and private hospitals, specialists, and subspecialist clinics.^[31,33]

National Health Insurance

To achieve UHC for the population, Indonesia launched the NHI scheme, known as *Jaminan Kesehatan Nasional (JKN)*, in 2014. The *JKN* aims to provide basic health services that covers various aspects of healthcare to all Indonesians.^[34] The *JKN* scheme has enrolled approximately 250 million people, making it one of the largest health insurance schemes in Indonesia.^[34] The health insurance budget is managed by *BPJS Kesehatan*, a single health purchasing agency, and it is financed by the government, employers, employees, and insured individuals.^[30] The primary healthcare payment system operates on a capitation basis, where general dentists receive payment for specific health services provided to each registered patient within a defined period. Oral examination, basic treatments,

and emergency are covered under this scheme, with advanced treatments referred to specialized healthcare facilities [Supplementary Figure 2].

Nusantara Sehat program

The government launched a voluntary-based program “*Nusantara Sehat*” in 2015 to address the shortage of health personnel in remote areas, borderlands, and areas with certain health problems.^[23] Health personnel apply and undergo a selection process before working in designated districts. They are assigned to community health centers or hospitals where they work collaboratively with other health personnel. Registration for the *Nusantara Sehat* program is open biannually, and the health personnel will be distributed nationwide, with a focus on remote regions that lack dental healthcare professionals. The project typically spans 2 years, which is extendable upon the personnel’s willingness to continue service in the targeted area.

LIFE-COURSE ORAL HEALTH ORAL HEALTH CARE PROGRAM

In assuring oral health care for the population, the government set particular programs, referring to the 2015 Regulation of Minister of Health No. 89

[Supplementary Table 4].^[35] They attempted to provide treatment for everyone at each life stage, from the gestation phase (expectant mothers) to the aging population.

Oral health services for pregnant women commence concurrently with the first antenatal check, where pregnant women are provided with an “Oral Health Scoring Card” to assess their periodontal status and the presence of dental caries every trimester.^[35,36] Additionally, pregnant women receive oral health counseling and education from the dental team, along with referrals for further curative treatment if necessary.

Under Regulation No. 89, issued by the Indonesian Minister of Health in 2015, oral healthcare programs target infants, toddlers, and preschoolers, implemented across various settings such as healthcare facilities, daycare centers, and educational institutions.^[37] Basic dental prevention and treatment are covered under the NHI scheme. The *Usaha Kesehatan Gigi Sekolah (UKGS)* program offers integrated dental healthcare services specifically for elementary and middle school-aged children, including dental and oral health screening, education, regular checkups, mass tooth-brushing initiatives, and advanced dental services.^[38]

Oral health services for older adults prioritize curative and rehabilitative approaches, aiming to restore masticatory function.^[35] “Integrated Health Services Posts” have been established in designated areas to improve access to healthcare for older people, driven by the community and implemented through primary healthcare centers (*Puskesmas*). This program emphasizes integrating comprehensive health services for older individuals, including oral health prevention, maintenance, behavior management, and training for cadres and gerodontologists.^[39]

Oral healthcare is also crucial for those with special needs and disabilities. Modified treatments are adapted to their developmental phases based on each specific type and severity condition, as well as any associated complications. The direction also emphasized on the multidisciplinary approach and educational intervention for the caregivers for the provision of oral health support.^[36]

DISCUSSION

To the best of our knowledge, this study represents the first comprehensive review of Indonesia’s oral health status and oral healthcare system after the NHI implementation. The most recent National Basic Health

Survey in 2018 (*Riskesdas*) revealed unsatisfying results regarding the oral health status of the population.^[9] Despite common oral health issues shared with other countries, Indonesia faces unique challenges in its efforts to improve oral health conditions. This necessitates strengthening oral health conditions, strengthening health systems, and establishing policies addressing oral health needs comprehensively.

According to the National Health Action Plan of Indonesia, the government set the target to achieve the DMF-T score for the 12-year group, that is, 1.14 in 2025, and the long-term target is set to achieve DMF-T 1 in 2030.^[40] This target sounds ambitious without any significant effort since the current DMF-T is still exceeding the target, which was around 1.9 at the national level and varied between 1.6 and 3.2 in some previous studies [Supplementary Table 1].^[9,10,13,14,17,18,41,42] This is similar to the prevalence of periodontal disease, which is still high, that is, around 19.6% at the national level, and varied between 28% and 100% among different age groups [Supplementary Table 2].^[43]

Several possible reasons for the oral health status of the underprivileged Indonesian population suggest inadequate access to dental care. The country’s unique topography may also be contributing to these challenges. Further efforts are needed to sustainably improve access to dental care, with the Ministry of Health taking responsibility and receiving support from other governmental and non-governmental organizations.

The role of oral health personnel is crucial; certain regions in Indonesia experience a shortage of dental professionals, resulting in a heavier workload for available health workers. According to the WHO standards, the number of oral health personnel falls below the recommended level, with only 1.2 dentists per 10,000 population in 2021.^[44] This is notably lower than the statistics in neighboring countries such as Malaysia (2.9:10,000 in 2018), Thailand (2.5:10,000 in 2019), and the Philippines (2.5:10,000 in 2021).^[45] Given the scarcity of dentists in Indonesia, dental therapists are vital in oral healthcare prevention and promotion. However, the uneven distribution of the dentist workforce and dental therapists nationwide results in untreated oral health issues and neglect of preventive measures, particularly in remote areas. To mitigate the shortage of dental personnel for providing preventive care, it may be necessary to introduce people from additional oral healthcare professions such as dental hygiene. Furthermore, national-level implementation of oral health policies is essential to improve oral healthcare services, considering the available dental workforce.

Addressing healthcare expenditure is crucial. Indonesia's healthcare facility budget has remained stagnant, hovering between 2.8% and 3.02% of GDP from 2013 to 2019.^[46] Although the government has spent more on healthcare-related expenditures in 2023 by about 5.3%, the allocation was mainly to tackle the post-pandemic situation, while also providing for sustaining the NHI scheme. Compared to neighboring countries, Indonesia's healthcare expenditures are lower than those of other Southeast Asian countries [Supplementary Figure 3].^[46] These financial challenges necessitate thorough assessment, especially considering the ongoing enrollment under the NHI scheme since 2014 and the persistent issue of excessive out-of-pocket expenses despite limited improvement in national health expenditures.

While the NHI scheme aims to provide comprehensive healthcare to Indonesians, gaps in coverage or accessibility may exist, particularly for rural populations. Despite the implementation of the *Nusantara Sehat* program in 2015 to address this issue, significant improvements have yet to be observed.^[23] Efforts should be made to ensure equitable access to dental care services for all Indonesians, regardless of the location or socioeconomic status. This may involve expanding the network of covered healthcare facilities and improving the infrastructure in underserved areas. Additionally, utilizing digital or teledentistry to address these issues may be feasible with available technological resources and internet coverage.

Despite oral healthcare services tailored to various life stages, accessibility remains challenging. While programs prioritize preventive approaches, gaps in implementation exist. Emphasizing preventive measures such as regular checkups, oral hygiene education, and early intervention strategies could help mitigate oral health problems. Programs in Indonesia target specific age groups, but there is a lack of emphasis on early childhood care.

This review has some limitations. The lack of recent oral health survey data, unavailable since 2018, significantly restricts the analysis. Consequently, the research findings may not fully capture the country's oral health status since most regulations governing the national oral healthcare system are managed within the broader healthcare framework. The absence of comprehensive evaluation reports on oral health systems and prevention programs makes it challenging to evaluate the successful rate of the programs. Nevertheless, this review offers essential and updated insights to inform better oral health policies and set specific goals for oral health

promotion in Indonesia. There is a need to strengthen access to oral healthcare services, particularly in remote areas, through collaborative efforts in Indonesia. By prioritizing oral healthcare within national health agendas and implementing evidence-based strategies, Indonesia can work toward achieving better oral health outcomes for its population.

In conclusion, the high prevalence of dental caries and periodontal diseases, as well as DMF-T, which is considerably higher than the national target, suggests that the oral health status of Indonesian people remains dissatisfying. To overcome that situation, strengthening primary care and its ongoing programs, prioritizing oral health with emphasis on preventive measures in the national agenda, enhancing workforce capacity, and promoting preventive measures are crucial steps for future directions.

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CONFLICTS OF INTEREST

Nil.

AUTHORS CONTRIBUTIONS

Not applicable.

ETHICAL POLICY AND INSTITUTIONAL REVIEW BOARD STATEMENT

Not applicable.

PATIENT DECLARATION OF CONSENT

Not applicable.

DATA AVAILABILITY STATEMENT

The data set used in the current study is data availability within the article or its supplementary materials.

Abbreviations

CPITN – Community periodontal index of treatment needs

DMFT/dmft – Decayed, missing, filled teeth (dmft: for deciduous teeth)

GDP – Gross domestic product
 JKN – *Jaminan Kesehatan Nasional* (national health security)
 MOH – Ministry of Health
 NHI – National Health Insurance
 PUSKESMAS – *Pusat Kesehatan Masyarakat* (Public Health Center)
 RISKESDAS – *Riset Kesehatan Dasar* (Basic Health Research)
 UHC – Universal health coverage
 UKGS – *Usaha Kesehatan Gigi Sekolah* (school-based oral health program)
 WHO – World Health Organization

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Supplementary Table 1: DMFT/dmft Index from National Health Survey 2013 and 2018

Age groups (years)	2013				2018			
	DT/dt	MT/mt	FT/ft	DMFT/dmft	DT/dt	MT/mt	FT/ft	DMFT/dmft
3–4*	-	-	-	-	-	-	-	6.2
5*	-	-	-	-	-	-	-	8.1
12	1.0	0.3	0.0	1.4	1.8	0.1	0.0	1.9
15	1.1	0.3	0.0	1.5	2.3	0.0	0.0	2.4
18	1.1	0.4	0.1	1.6	-	-	-	-
35–44	2.0	3.3	0.1	5.4	5.2	1.7	0.1	6.9
45–54	2.1	5.6	0.1	7.9	5.9	3.2	0.1	9.2
55–64	2.1	10.1	0.1	12.3	6.6	5.9	0.1	12.6
65+	1.8	17.0	0.1	18.9	6.6	10.1	0.0	16.8

Data resources were from the National Basic Health Survey in 2013 and 2018.

*Primary teeth. DMFT = decayed, missing, filled permanent teeth, dmft = decayed, missing, filled primary teeth, DT/dt = decayed permanent/primary teeth, MT/mt = missing permanent/primary teeth, FT/ft = filled permanent/primary teeth, - = not available

Supplementary Table 2: Number of dental health workforces

	2018	2019	2020	2021
Dentists	27,860	33,135	34,837	33,652
Specialist dentists	3588	4184	4500	4483
Dental therapists	-	18,003	-	19,600

- = not available

Supplementary Table 3: National Health Insurance (NHI) schemes

Characteristics	NHI schemes		
Targeted population	Underprivileged populations	Formal public and private sector	Informal sector
Number of enrolled	252.1 million (March 2023)		
Contributors	Government 100%	5% of salary	Member 100%
Contribution rate	IDR 42,000 (\$ 2.7) per member per month	Public sector: employee 2%; employer 3%. Private sector: employee 1%; employer 4%.	By class (per member per month): Class I → IDR 150,000 (\$ 9.6) Class II → IDR 100,000 (\$ 6.5) Class III → IDR 35,000 (\$ 2.3)
Benefits	Health promotion; preventive, curative, and rehabilitative treatments; medically indicated lab tests, drugs, and supplies; and ambulance service for referrals		
Coverage for dependents	Subsidized: per person	Spouse + children under 21 or under 25 if still studying	Per person basis

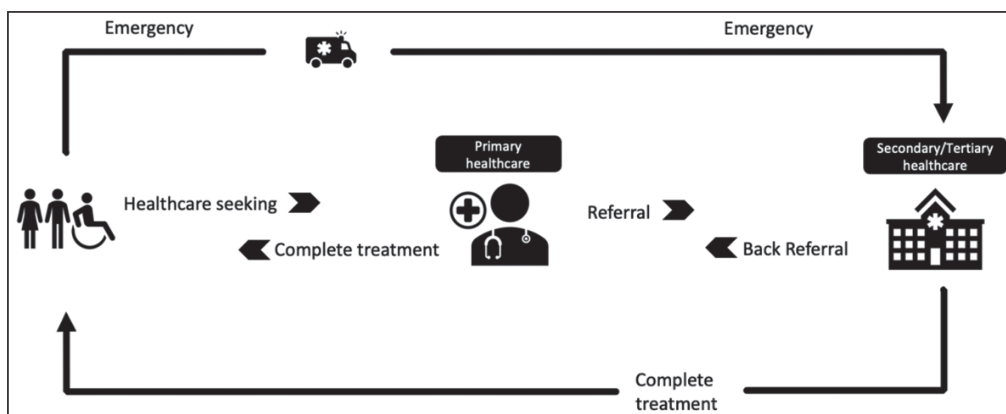
IDR = Indonesian Rupiah, \$ = United States Dollar

Supplementary Table 4: Life-course Oral Healthcare Services (Ministry of Health in Indonesia, 2015)

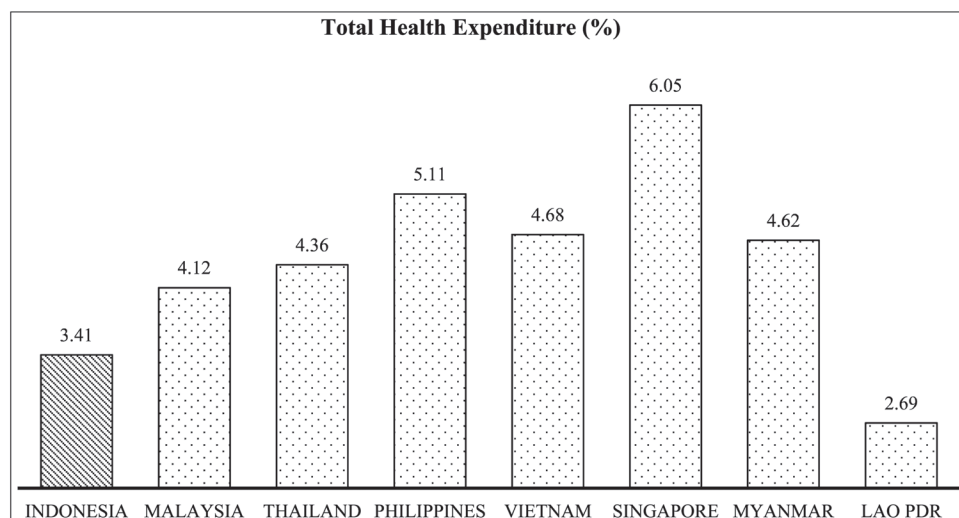
	Pregnant women	Children and adolescents	Older adults	Individuals with special needs
Target	1.Pregnant women 2.Fetus	1.Infants 2.Toddlers 3.Pre-school children 4.Primary school children 5.Middle school children	Older adults (≥60 years)	People with special needs
Details	This kind of dental care aims to maintain and improve the dental and oral health of pregnant women to help optimize the mother's overall health for good fetal growth and development and to prevent congenital diseases, particularly in the oro-facial region.	This kind of dental care focuses on promotive and preventive approaches without neglecting curative and rehabilitative care.	Dental and oral healthcare services for the older adults are prioritized on services with a curative and rehabilitative approach in the form of treatment and restoration of masticatory function according to dental and oral health problems in the older adults.	Dental and oral health services for individuals with special needs are carried out in accordance with the developmental phases of individuals with disabilities and/or treatment modifications according to the type of disability, level of disability, and whether there are any complications.
Location	Primary healthcare center (<i>Puskemas</i>)	Primary healthcare center (<i>Puskemas</i>), Integrated Healthcare Center (<i>Posyandu</i>), preschools group, and schools.	Primary healthcare centers (<i>Puskemas</i>) and other places provide oral healthcare for the community.	Health service facilities that provide dental and oral health services are equipped with special facilities and infrastructure for dental and oral care for persons with disabilities. Specialist dentists, general dentists, and dental therapists.
Service provider	The dentist or primary healthcare center's cadres.	Medical personnel, primary healthcare center's cadres, teachers, and caregivers.		Specialist dentists, general dentists, and dental therapists.
Activities	Dental and oral health services for pregnant women are carried out integrated with the antenatal examination of pregnant women since the first antenatal visit (K1) in the primary healthcare center (<i>Puskemas</i>). 1.Oral health counseling. 2.Oral health screening and early detection. 3.Referral to the specialist dentist for further curative treatment indications.	1.For infants, toddlers, and pre-school children, dental care is given to the parents or caregivers, such as dental counseling and oral hygiene training. 2.For primary and middle school children, oral health promotion and prevention are included in <i>Usaha Kesehatan Gigi Sekolah</i> (UKGS) program. This program supports oral health screening, dental education, regular dental health checkup, and advanced dental care.		1.Oral health counseling. 2.Oral healthcare training for caregivers or supporters. 3.Dental treatments.



Supplementary Figure 1: Distribution of dentists in healthcare facilities



Supplementary Figure 2: Flow service using the NHI scheme



Supplementary Figure 3: Comparison of health expenditure in the Southeast Asia region as of 2020